MICHAEL F. PÉREZ

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My research focuses on video understanding, including action recognition, video segmentation, and video retrieval. I develop and evaluate learning-based algorithms and scalable, production-oriented pipelines for representing, indexing, and retrieving video content, with an emphasis on efficiency, scalability, and real-time inference. My work bridges core machine learning research and robust systems through end-to-end experimentation and engineering.

EMPLOYMENT

Senior Engineer August 2026 – Present
Raft Tampa, FL

Machine Learning Engineer Intern May – August 2025
CoVar Durham, NC

- Improved resolution of infrared video via open-source super-resolution model to classify distant objects.
- Developed and trained lightweight RGB object-detection models for real-time drone detection, targeting resource-constrained, on-device use.
- Built offline computer-vision evaluation pipelines spanning preprocessing, model inference, metric computation, and efficiency analysis.

Research Assistant August 2020 – August 2026
University of Florida Gainesville, FL

- Member of Ruiz Human-Computer Interaction Lab, Data Science Research Lab, and Graphics, Imaging and Light Measurement Lab.
- Conducted interdisciplinary research applying computer vision to AR, web, and neuroscience projects.
- Principal Investigators: Dr. Jaime Ruiz, Dr. Daisy (Zhe) Wang, Dr. Corey Toler-Franklin.

Teaching Assistant August 2020 – August 2026 (Various Semesters)
University of Florida Gainesville, FL

- TA for Analysis of Algorithms, Data Structures and Algorithms, Computational Structures for Computer Graphics, Operating Systems, Programming Fundamentals II, Performant Programming in Python, Deep Learning for Computer Graphics, Algorithm Abstraction and Design
- Graded assignments, developed automatic grading scripts, hosted discussions, and held office hours for 100s of students.

Research Assistant, Research Intern June 2019 – May 2020
Florida Polytechnic University Lakeland, FL

- Applied computational methods to rhinoplasty research.
- Principal Investigator: Dr. Oguzhan Topsakal.

Student Educational Assistant June 2019 – May 2020
Databases 1 and Software Engineering, Florida Polytechnic University Lakeland, FL

- Graded 50 assignments and 50 quizzes biweekly for Database 1 and Software Engineering.

EDUCATION

Ph.D. in Computer Science August 2026
University of Florida, GPA: 3.77 Gainesville, FL

M.S. in Computer Science May 2023
University of Florida, GPA: 3.75 Gainesville, FL

B.S. in Computer Science May 2020
Florida Polytechnic University, GPA: 3.8 Lakeland, FL

TECHNICAL SKILLS

***Applications*:** Git, LaTeX, Visual Studio, CMake, MATLAB, Blender, Docker
***Programming Languages*:** Python, JavaScript, C++, C, HTML, CSS, GLSL, Java, Julia, C#, SQL
***Libraries*:** PyTorch, OpenCV, Numpy, SciPy, Pandas, Tensorflow, SciKit-Learn, OpenGL, WebGL, FastAPI, Flask, ROS, Hugging Face
***Operating Systems*:** Ubuntu, Mac OS, Windows
***Spoken Languages*:** Bilingual (English and Spanish)

RESEARCH EXPERIENCE

Ph.D. Thesis: Efficient Video Action Detection 2025 – Present
CISE Department, University of Florida Gainesville, FL

- Developing efficient action-detection models for higher temporal accuracy and lower latency in long videos.
- Built a unified framework to benchmark video action-detection models across mAP, latency, throughput, memory, and FLOPs.
- Preparing a first-author WACV 2027 manuscript.

ECOLE Program - CReLeRI Team (DARPA-funded) July 2023 – May 2025
Ruiz HCI Lab and DSR Lab, CISE Department, University of Florida Gainesville, FL

- Developed and integrated UI and image/video analysis APIs and trained/tested action segmentation models.

PTG Program - ENKIx Team (DARPA-funded) July 2023 – December 2024
Ruiz HCI Lab, CISE Department, University of Florida Gainesville, FL

- Trained and tested action recognition models for an AR task guidance system, achieved 34% F1-score in task recognition (mid-range results among DARPA performers), and conducted usability and cognitive load evaluations.

Animal Behavior Recognition (NSF-funded) August 2020 – July 2023
Graphics, Imaging, and Light Measurement Lab, CISE Department, UF Gainesville, FL

- Investigated adversarial approaches for modeling 3D motion and behavior in videos for applications in biomedical research.
- Conducted a survey of animal action recognition algorithms, trained and tested existing SOTA methods, and began implementing novel semi-supervised adversarial algorithms.

Rhinoplasty Research Group May 2019 – May 2020
Computer Science Department, Florida Polytechnic University Lakeland, FL

- Conducted a survey of over fifty research papers, documenting the facial landmarks and measurements relevant to the field of rhinoplasty.
- Collaborated with other students to develop a web interface for importing, exporting, and displaying feature point coordinates and measurements values onto 3D models of the human face.
- Achieved 95% agreement with Blender software and outperformed 2D methods in inter-/intra-rater reliability and usability metrics.

HONORS AND AWARDS

GenerationNext Grant — January 2021 – August 2026

- Multifaceted cohort-based support system to aid Ph.D. students in the UF CISE department.

Graduate School Preeminence Award — September 2020 – August 2026

- Highly competitive research assistantship stipend offered to UF CISE's strongest Ph.D. applicants.

L3Harris Corporation Graduate Fellowship — April 2022

- Scholarship awarded to five UF CISE graduate students based on academic performance (e.g., GPA), research activity (e.g., publications and awards), and professional services.

PROFESSIONAL ACTIVITIES & SERVICE

Peer Reviewer — August 2026
ICXR 2026

Peer Reviewer — March 2026
ACM International Conference on Multimedia Retrieval (ICMR)

Peer Reviewer — May – June 2025
ACM Multimedia

Co-Chair — Spring 2025
UF Human-Centered Data Science Seminar

Peer Reviewer — October – November 2024
IEEE Conference on Virtual Reality and 3D User Interfaces

Panelist — February 2023
Black Male Computer Scientist Panel**,** *Reichert House Youth Academy* — Gainesville, FL

- Engaged with high schoolers to provide college and career advice.

PUBLICATIONS

Journal Articles

1. R. Shahriari, Y. Yang, D. Tamboli, M. F. Pérez, Y. Zha, J. Hou, M. Deng, E. D. Ragan, J. Ruiz, D. Z. Wang, Z. Hu, and E. Xing, "**MuCHEx: A Multimodal Conversational Debugging Tool for Interactive Visual Exploration of Hierarchical Object Classification,**" *IEEE Computer Graphics and Applications*, no. 1, pp. 1-13, 2025. doi: 10.1109/MCG.2025.3598204.
2. M. M. Celikoyar, M. F. Pérez, M. I. Akbaş, and O. Topsakal, "**Facial Surface Anthropometric Features and Measurements with an Emphasis on Rhinoplasty**," *Aesthetic Surgery Journal*, vol. 42, no. 2, pp. 133–148, 2021. doi: 10.1093/asj/sjab190.
3. O. Topsakal, M. İ. Akbaş, B. S. Smith, M. F. Pérez, E. C. Guden, and M. M. Celikoyar, "**Evaluating the agreement and reliability of a web-based facial analysis tool for rhinoplasty**," *International Journal of Computer Assisted Radiology and Surgery*, vol. 16, no. 8, pp. 1381–1391, 2021. doi: 10.1007/s11548-021-02423-z.
4. O. Topsakal, M. İ. Akbaş, D. Demirel, R. Nunez, B. S. Smith, M. F. Pérez, and M. M. Celikoyar, "**Digitizing rhinoplasty: A web application with three-dimension preoperative evaluation to assist rhinoplasty surgeons with Surgical Planning**," *International Journal of Computer Assisted Radiology and Surgery*, vol. 15, no. 11, pp. 1941–1950, 2020. doi: 10.1007/s11548-020-02251-7.

Conference Proceedings

1. Michael Pérez, Danish Tamboli, Haodi Ma, Reza Shahriari, Vyom Pathak, Dzmitry Kasinets, Daisy (Zhe) Wang, Jaime Ruiz, Eric Ragan, Yichi Yang, Yuheng Zha, Enza Ma, Zhiting Hu, Eric Xing, and Jun-Yan Zhu. "**CReLeRI: Explainable, Concept-centric, Representation, Learning, Reasoning, and Interaction Video Analysis System**," *ACM International Conference on Multimedia*, 2025. doi: 10.1145/3746027.3754479.
2. M. F. Pérez, P. Torrione, and D. Kalika, "**Super-Resolution for Improving Vehicle Recognition at Large Stand-offs**," in *Military Sensing Symposium (MSS) (BSD, Materials & Detectors, and Passive Sensors) Conference*, March 2-6, 2026.
3. O. Topsakal, M. İ. Akbaş, B. S. Smith, M. F. Pérez, E. C. Guden, and M. M. Celikoyar, "**Evaluating Intra and Inter Reliability of a Web-Based Facial Analysis Tool for Rhinoplasty**," in *Computer Assisted Radiology and Surgery (CARS) Conference*, Munich, Germany, June 21-25, 2021.
4. O. Topsakal, M. İ. Akbaş, D. Demirel, R. Nunez, B. S. Smith, M. F. Pérez, and M. M. Celikoyar, "**Digitizing Rhinoplasty: A web application for three-dimensional preoperative planning**," in *Computer Assisted Radiology and Surgery (CARS) Congress*, Munich, Germany, June 23-27, 2020.

Manuscripts in Preparation

1. M. F. Pérez, A. Barquero, R. Calvo, D. Moore, M. Zolotas, N. Grimaldi, Y. Liu, E. Patterson, D. Bista, N. Regmi, R. Venkatakrishnan, L. Anthony, K. Brown, D. Wozniak, I. Wang, A. Tompkins, J. Fairbanks, T. Padir, M. Zahabi, D. Kaber, and J. Ruiz. "**ENKIx: An Intelligent AR Task Guidance System with Cognitive Load-Adaptive Capabilities**," *IEEE Transactions on Human-Machine Systems*, 2026.
2. M. F. Pérez, Daisy (Zhe) Wang, and Jaime Ruiz, "**Temporal Granularity as a Design Variable in Zero-Shot Video Retrieval**", *International Conference on Multimedia Retrieval*, 2026.
3. R. Calvo, C. Bowers, M. F. Pérez, A. Barquero, and J. Ruiz, "**The Effects of Multiple Companion Agents on User Experience and Mental Workload**", *Manuscript in preparation*, 2026.

Preprints

1. M. F. Pérez and C. Toler-Franklin, "**CNN-Based Action Recognition and Pose Estimation for Classifying Animal Behavior from Videos: A Survey**," *arXiv preprint arXiv:2301.06187*, 2023.

REFERENCES

Dr. Jaime Ruiz, Associate Professor (Ph.D. co-supervisor)
Computer and Information Science and Engineering Department
University of Florida, Gainesville, FL
jaime.ruiz@ufl.edu

Dr. Daisy (Zhe) Wang, Professor (Ph.D. co-supervisor)

Computer and Information Science and Engineering Department
University of Florida, Gainesville, FL
daisyw@cise.ufl.edu

Dr. Corey Toler-Franklin, Assistant Professor (former Ph.D. supervisor)
Computer Science Department
Barnard College, New York, NY
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Dr. Oguzhan Topsakal, Associate Professor (undergraduate research supervisor)
Computer Science Department
University of South Florida, Tampa, FL
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